Same Trajectories, Different Rankings: Success Timing and Protocol Choices in Latent World-Model Planning

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Background: Recent latent-world-model papers report success on the four-task LeWM benchmark, usually counting an episode as successful when a trajectory first enters a target region.
Objectives: We measure how success timing, observation history, planner settings, and sampling choices change the reported number and the method ranking.
Methods: Public checkpoints and released code were evaluated on paired cases with three fixed planner random-seed streams. We changed one protocol choice at a time and applied any-step, block-boundary, final-step, and sustained-window labels to the same 50-step trajectories. The three preregistered predictions concern any-step success and were tested on 100 held-out cases per task; endpoint and sustained-window comparisons re-score the same trajectories.
Results: Three-frame history raises Reacher any-step success by +12.7 percentage points and final-step success by +48.3 [+42.0, +55.0] points. Yet three-frame LeWM is successful at the final step in 85.7% of runs and at all ten final environment-step endpoints in only 0.3%; endpoint occupancy is not sustained occupancy. AP and AD-WM retain common-protocol gains of +15.8 to +27.0 points on TwoRoom and Cube. A corrected continuation adapter passes checks on all 2700 saved Reacher trajectories; our earlier adapter had allowed an unintended internal reset after success. Corrected comparisons show reversals for the same controller pair across arrival, endpoint, and sustained-window labels. In Reacher, three-frame LeWM minus AP is -2.7 [-4.7, -1.0] points for any-step, +19.7 [+12.3, +27.3] for final-step, and -10.3 [-14.3, -6.7] for all-last-ten success. In PushT, AP minus one-frame LeWM is +0.7 [-5.3, +6.7], -21.7 [-31.3, -11.7], and +25.3 [+17.3, +33.3] points, respectively: the final-step and sustained-window intervals lie on opposite sides of zero.
Conclusions: First arrival, occupancy at designated endpoints, and sustained-window occupancy answer different control questions and can rank the same controllers differently. A benchmark result requires explicit check times, continuation semantics, history construction, planner budget, and paired sampling.

Additional Key Words and Phrases: world models, model predictive control, evaluation protocols, reproducibility, benchmark auditing

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1 Introduction

A continuous held-out Reacher trajectory reaches the joint target at an environment-step endpoint at step 16, leaves it by step 18, and is outside the target at step 50. Figure 1 shows the two joint-error curves from this corrected rollout. First-arrival success asks whether a controller reaches the target within the budget. Endpoint occupancy asks whether it is there at designated block boundaries or at the final step. Sustained-window occupancy asks

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how much, or whether all, of the final window is spent inside. These three questions produce different rankings from the same continuous trajectories.

Observation-and-action history has a large task-specific effect. In Reacher, three-frame LeWM gains +12.7 points in any-step success and +48.3 [+42.0, +55.0] points at the final step relative to one-frame LeWM. Its final-step rate is 85.7%, but all-ten-step success is only 0.3%. AP instead has 66.0% final-step success, 57.4% mean last-ten occupancy, and 10.7% all-ten success. Thus endpoint improvement does not imply sustained holding (Table 11).

The same controller pair can reverse order across arrival, endpoint, and sustained-window labels. For three-frame LeWM minus AP in Reacher, any-step, final-step, and all-last-ten contrasts are -2.7 [-4.7, -1.0], +19.7 [+12.3, +27.3], and -10.3 [-14.3, -6.7] points; all three intervals exclude zero, with two opposite orderings. For AP minus one-frame LeWM in PushT, the corresponding contrasts are +0.7 [-5.3, +6.7], -21.7 [-31.3, -11.7], and +25.3 [+17.3, +33.3] points. Any-step success is unresolved, while final-step and sustained-window success establish opposite orderings.

Common-protocol gains remain large for AP and AD-WM on TwoRoom and Cube. Interpreting post-success behavior also requires a continuous physical rollout. Continuing a Reacher rollout past success requires disabling the task's success termination. Our earlier continuation adapter cleared only the outer `terminated` flag; dm_control had already scheduled a reset, so the arm was re-randomized on the next step. The corrected adapter preserves every saved any-step outcome in all 2700 formal records.

This distinction matters because the LeWM benchmark is shared by several latent-planning papers, while the published descriptions expose different parts of the evaluation protocol. Our selected inventory contains 12 sources with results on at least one of the four tasks and 8 sources reporting all four (search cutoff: 2026-10-12 UTC). The records vary in how they specify success thresholds, early termination, observation history, CEM iterations, case counts, and planning seeds. These fields determine what a percentage measures and how finely a comparison can be resolved.

We ask three questions. Which choices move a measured success number? Which method differences survive when cases, streams, and planner settings are aligned? Does changing the success definition change the ranking? We answer them with a paired read-only evaluation of public checkpoints and code. Each case is run with three fixed planning streams, protocol factors are changed one at a time, and the same full-budget trajectories support first-arrival, designated-endpoint, and sustained-window measurements. The original any-step predictions use held-out confirmation.

The remaining question is empirical precision. Three planning streams over 100 cases have a typical paired half-width of 4.83 points in Reacher and 1.67 points in PushT under the earlier any-step noise analysis. The continuous-run audit, method alignment, and uncertainty measurements together identify which evaluation choices change conclusions.

The paper contributes a paired evaluation design that exposes these protocol effects, a reproduction table with measured intervals, a full-trajectory comparison of arrival, endpoint, and sustained-window labels, a literature protocol inventory, and a reporting checklist. Section 2 introduces the planner and tasks; Section 3 describes the audit; Sections 4–8 present reproduction, factor effects, common-protocol gains, success-label reversals, and resolution; Sections 9–10 discuss implications and conclusions. The appendices provide the timeline, complete tables, planner-iteration analysis, compute accounting, and a reproducibility checklist.

2 Background and Related Work

2.1 Latent world-model planning

A latent world model maps an observation and an action into a learned representation and predicts future representations without reconstructing every pixel. A model-predictive-control (MPC) planner samples candidate

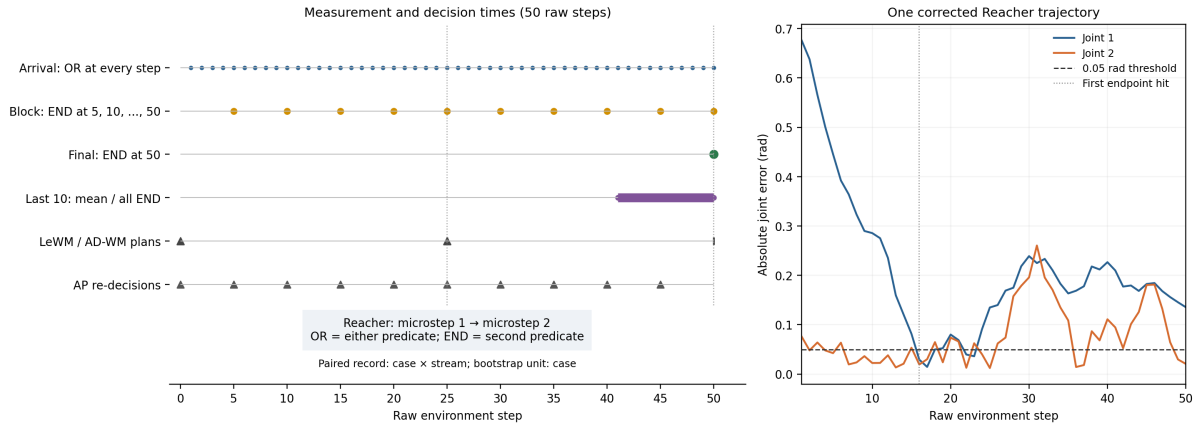


Fig. 1. Measurement times and an example trajectory. A case-stream identifies a paired trajectory; the case is the bootstrap unit. The right panel shows a corrected continuous Reacher trajectory, its two absolute joint errors and the 0.05 rad threshold. The left panel marks block checks, 25/50-step plan endpoints, AP re-decisions, window reads, and the Reacher OR/END microstep distinction. The right panel is the recorded corrected trajectory.

action sequences, predicts their consequences, scores the candidates against a goal, and executes a prefix before planning again. The cross-entropy method (CEM) narrows the candidate distribution by repeatedly retaining an elite set and refitting its mean and variance. The LeWM evaluation executes a complete 25-step plan before the next query in its reference configuration; alternative released planners shorten the executed prefix or alter the goal score. Figure 1 depicts the objects that are held fixed and the labels applied after a full trajectory has been produced.

This paper evaluates the original LeWM checkpoint, Fast-LeWM, AP (Anchored Planning), and AD-WM. Fast-LeWM uses a one-frame input. AP anchors short plans to a frozen world model and retrieval bank. AD-WM changes the action-conditioned objective and reports three training seeds. These methods share the task interface but expose different planner choices, which makes protocol alignment necessary before comparing their percentages (Gao and Xu 2026; X. Liu et al. 2026; Maes, Le Lidec, Scieur, et al. 2026; Qiu et al. 2026).

2.2 Tasks and success tests

The four tasks cover a two-joint arm, a planar pusher, a point agent crossing two rooms, and a three-dimensional cube manipulation environment. The audit uses the task thresholds in the released interfaces; no threshold is changed. Table 1 summarizes the tests and the common 50-step budget.

2.3 Published protocols and released configurations

LeWM Appendix D describes 30 CEM iterations for PushT and 10 for the other tasks; Appendix B states 30 without a task qualification. The released evaluation configuration uses 30 throughout. The appendix lists three history frames for PushT and Cube, one for TwoRoom, and does not state a Reacher value; the library default and the released configuration supply one frame. Case counts and planning seeds also differ across sources. Table 2 records the value or “not specified” for each source rather than filling gaps by inference.

The selected inventory contains 12 sources, including benchmark papers, planner papers, and objective studies. 8 cover all four tasks. We included a source when it reported a result, protocol, or released configuration for at least one benchmark task and was publicly identifiable by the 2026-10-12 UTC cutoff; the table is an inventory of

Table 1. The four benchmark tasks and their success tests.

Task	Controlled system	Success test	Budget/offset
Reacher	Two-joint arm	Both joint-angle errors are below 0.05 rad.	50 steps; native goal
PushT	Pusher and T-shaped block	Planar (x, y) position error of the pusher/block state is below 20 pixels and the block angle error is below $\pi/9$.	50 steps; native goal
TwoRoom	Point agent in two rooms	Euclidean image distance below 16 pixels.	50 steps; native goal
Cube	3-D single-block manipulation (OGBench Cube-v0)	Euclidean distance between the block position and target is $\leq$ 0.04 m; target quaternion is not part of the predicate.	50 steps; native goal

Table 2. Selected inventory of published protocols and released configurations (search cutoff 2026-10-12 UTC). "paper", "config", "code", and "not stated" identify the strongest evidence source for each compact field. History means planner input frames; NS = not stated. Seeds are split by category when identifiable. P/C/T/R denote PushT/Cube/TwoRoom/Reacher; rel. means released. The 12 displayed sources are the complete selected table; eight explicitly cover all four benchmark tasks. DeepJEPA's five entries include Cube variants and omit TwoRoom.

Source	Tasks	Success	Early stop	History	CEM iterations	Cases	Seeds	Evidence
LeWorldModel (Maes, Le Lidec, Scieur, et al. 2026)	4	stated	yes	3 P/C; 1 T; R NS	30 P; 10 C/T/R (App. D; App. B states 30 without qualification); 30 all (rel. config)	100	3 planner	paper/config
Fast LeWorldModel (Gao and Xu 2026)	4	NS	NS	1	30	50	7 training	config/paper
Anchored Planning / AP (X. Liu et al. 2026)	4	stated	yes	1	30	128	3 planner	paper/code
AD-WM (Qiu et al. 2026)	4	NS	NS	1	30	50	3 training	paper/config
PPWM (Feng et al. 2026)	4	any/terminal	NS	NS	NS	100	3 planner	paper
ProWorld (Z. Liu et al. 2026)	4	stated	NS	NS	NS	NS	NS	paper
Singh reproduction (Singh 2026a)	1 (TwoRoom)	stated	NS	NS	30	50	NS	paper/code
Objective bottleneck / Singh (Singh 2026b)	1	NS	NS	NS	30	50	NS	paper
Temporal-Distance JEPA (Bai and Xiong 2026)	4	NS	NS	5	30 (most)	50	1 training	paper/config
DeepJEPA (Jin et al. 2026)	5 (Cube variants)	NS	NS	NS	30	50	3 training	paper
Decision-Metric Alignment (Wang et al. 2026)	4	stated	NS	NS	30	50	3 training	paper
Keeping JEPA Plannable (Strohm et al. 2026)	1	stated	no	NS	30	50	NS	paper

selected sources. Evidence labels distinguish a value written in the paper (paper), determined from a released configuration (config), determined from source code (code), or not determinable (not stated). The protocol fields are not interchangeable: an unspecified history value cannot be treated as a reported three-frame setting, and a

paper's case count does not define the number of streams in a released configuration. We therefore reproduce source-specific numbers in Section 4 and use the reference protocol for aligned contrasts.

2.4 Related work

Evaluation methodology in reinforcement learning has documented the influence of random seeds, sample size, environment versions, and reporting summaries on conclusions (Agarwal et al. 2021; Gundersen et al. 2024; Henderson et al. 2018; Machado et al. 2018). Recent work on benchmark reporting and multi-task suites extends this concern to task definitions and environment wrappers (McLean et al. 2025; Patterson et al. 2024). Our audit instantiates those concerns for latent world-model planning by holding cases and planning streams paired within each comparison.

Several recent papers study the same benchmark. PPWM is the closest predecessor: it evaluates the same four tasks with LeWM, Fast-LeWM, PP-LeWM, and several planners, using 100 episodes, three planner seeds, and complete 50-step traces to report both anytime and terminal success; its Reacher table already shows high anytime and lower terminal rates and therefore contains the ingredients for metric-dependent rankings (Feng et al. 2026). Singh's independent reproduction is a direct protocol-audit predecessor: it reproduces the TwoRoom task and checks how the evaluation convention determines the result (Singh 2026a); a companion paper studies the objective bottleneck (Singh 2026b). AP's Appendix D.2 separately audits the LeWM configuration and evaluation protocol (X. Liu et al. 2026). Vesessook and Yang study planner scoring times and replanning intervals (Vesessook and Yang 2026); ProWorld discusses progress-aware scoring in long-horizon goal reaching (Z. Liu et al. 2026). AD-WM reports Cube results under its native setup and compares Fast-LeWM, Sub-JEPA, and INTACT public checkpoints (Qiu et al. 2026). We extend these lines with a protocol audit across later methods, fixed case-and-stream paired effects, preregistered held-out confirmation, and an explicit check of continuation semantics after success.

A second line of work changes the planning objective rather than the evaluation. TEMPO, TRM, and decision-metric alignment methods improve the match between a latent objective and the action that is eventually executed (Li et al. 2026; Moukheiber et al. 2026; Wang et al. 2026). In other settings, success definitions also change interpretation: dm_control's original Reacher task gives reward at each step spent inside the target, so episode return measures target occupancy more closely than first entry (Tassa et al. 2018). VIGIL contrasts a world state that is complete at episode end with an agent that correctly reports completion (Chen et al. 2026). These examples motivate recording check times and termination rules alongside a success percentage.

3 Audit Design

3.1 Checkpoints, code, and cases

We evaluated the public LeWM checkpoint, Fast-LeWM, AP, and AD-WM with the released adapters and task interfaces. The adapter and checkpoint identifiers were fixed before the formal runs. Fast-LeWM is evaluated with one-frame history because its released interface does not provide a three-frame condition. AD-WM's three training seeds are averaged within case and planning stream before contrasts are formed. Methods without public weights and code were inventoried but not substituted with private implementations.

The selected public methods were fixed by the earlier decision record: a public checkpoint and adapter available by the freeze date, a reproducible interface for at least one task, a distinct planner or objective choice, and a budget compatible with the paired audit. This is why LeWM, Fast-LeWM, Anchored Planning, and AD-WM are evaluated here. EpiWM and other public-weight candidates were considered in the earlier candidate inventory but are not part of the evaluated set when their released interface, checkpoint identity, or task coverage did not meet all four criteria.

Table 3. Case cohorts and exclusion levels. Each case is one window from one episode.

Cohort	Cases/task	Use	Exclusion level	AP retrieval bank
Original main	100	Source reproduction and first factor grid	Prior study cases not excluded from this cohort	Bank frozen before the first round
Fresh main	100	Common-protocol and factor sensitivity	Prior source-reproduction cases excluded	Bank rebuilt after prior evaluation episodes were removed
Held-out	100	Preregistered confirmation	Whole episodes used by this study's cases, refits, and retrieval bank excluded	Rebuilt bank excludes every evaluation episode, including held-out

The task interfaces come from the stable-worldmodel platform (Maes, Le Lidec, Facury, et al. 2026), audited at commit `63988116`. Success termination is a legitimate task design; the error was in our earlier post-success continuation adapter. The released first-arrival evaluation never continues past success and is unaffected. For Reacher, the corrected adapter preserves the released strict per-joint condition $|q_j - q_j^*| < 0.05$ rad but prevents success from terminating the underlying dm_control episode during the 50-step budget. Both native microsteps are executed for every environment step. We record the original predicate at each microstep, the two step labels, physical time, joint position and velocity, target configuration, and actual reset events. Appendix F reports the runtime checks and final-code technical gates.

The main case set contains 200 cases per task and the held-out case set contains 100 cases per task. Each case comes from a different episode and contributes one evaluation window; no episode contributes two cases to the same cohort. The original-main and fresh-main cohorts are each 100 cases per task. The held-out set excludes whole episodes used by this study's earlier evaluations, retrieval bank, and episodes used in the author's earlier retraining experiments on the same benchmark (those training results are not reported here). The case list is sampled from the benchmark pool with the recorded target-distance/start-state selection procedure, rather than selected after inspecting success. This exclusion prevents analysis choices in the study from selecting the held-out episodes.

3.2 Native settings and study interfaces

The public methods expose different planner interfaces. Table 4 maps each native setting to the common study interface. Native reset restores the task's released state variables (Reacher joint position/velocity, PushT state, TwoRoom proprioception, or Cube joint and block state); native goal uses the task's released goal state, with the Cube target reconstructed from the recorded target position and quaternion. Goal images are dataset endpoint images when the interface supplies them. Actions are converted through the released action normalisation and clipped at the environment bounds. The common study executes the same 50 raw environment steps and continues after an any-step success label; the corrected Reacher implementation prevents success from triggering an internal reset.

The planner and environment clocks use raw environment steps. A model prediction may cover a longer imagined horizon, but the released LeWM interface emits and executes a 25-step plan comprising five five-step action blocks, then queries again if budget remains. Steps 5, 10, …, 50 are action-block boundaries; 25 and 50 are endpoints of LeWM plans. AP re-decides every five raw steps and re-retrieves at that boundary. An

Table 4. Native planner settings mapped to the common study interface. All methods continue to 50 raw steps and use 300 CEM candidates, 30 elites, and 30 iterations. Actions are clipped at the environment bounds; LeWM, Fast-LeWM, and AD-WM use released normalisation, while AP uses its released adapter scaling. Fast-LeWM values come from released `config/eval/solver/cem.yaml`, task evaluation YAMLs and `eval.py`; the adapter packs five five-action blocks into its native long-prefix interface.

Method	Planned horizon / executed prefix	Re-decision / retrieval	Goal construction and CEM interface
LeWM	25-step predicted plan; five-step action block	Re-decision after five blocks (25 raw steps)	Released native goal and dataset endpoint image; released elite update
Fast-LeWM	Direct 25-raw-step action prefix (5 blocks × 5); all 25 executed	Re-decision every 25 raw steps; native horizon/receding horizon is 1 long block	Released native goal; CEM optimises one 25-action prefix
Anchored Planning	Short plan; five raw steps executed	Re-decision and retrieval every five raw steps	Frozen-world-model target with released retrieval bank; released update
AD-WM	LeWM-compatible plan; five-step action block	Re-decision after five blocks (25 raw steps)	Released native goal and dataset endpoint image; released update

action-block boundary is therefore distinct from a re-decision boundary: LeWM re-decides after 25 steps, whereas AP re-decides after five. Three-frame history means the current observation together with the real observations at $t-5$ and $t-10$ and the two intervening executed action blocks; the first query has only the current frame. No zero padding or predicted frames are inserted.

3.3 Paired streams and aggregation

A case is the statistical unit. Every method in a contrast receives the same case, the same reset, and one of three fixed planner random-seed streams. Pairing removes case difficulty from the first-order comparison and makes the interval describe the remaining case-level variation. For AD-WM, training seeds are averaged within case before the planning-stream average. All formal traces run the full 50-step budget even when an any-step label would already be true.

3.4 Protocol factors

The reference protocol uses native reset, one-frame history, 30 CEM iterations, 300 candidates, 30 elites, five five-step blocks, and a full 50-step trajectory. We change one interface at a time where a released implementation permits it. Figure 2 shows history length, the CEM elite standard-deviation correction, goal offset, goal image, and native replanning effects on the mixed 200-case set. The separate ten-versus-thirty iteration comparison uses a held-out set and appears in Appendix C.

3.5 Success labels and occupancy

Let e_t denote the task predicate at the end of environment step t, and let a_t denote whether it holds at any success check within that step. Reacher executes two microsteps, so a_t is their logical OR and e_t is the second microstep predicate; the other tasks use their released step-end check. Any-step success is $\mathbb{1}(\max_{1\le t\le 50} a_t = 1)$, block-boundary success is $\mathbb{1}(\max_{t\in\{5,10,\ldots,50\}} e_t = 1)$, and final-step success is e_{50}. Occupancy is $100 \times \Pr(e_t = 1)$, with training-checkpoint and planning-stream averaging inside each case. For the last K steps, mean occupancy is $K^{-1}\sum_{t=51-K}^{50} e_t$ and all-window success is $\prod_{t=51-K}^{50} e_t$, with $K \in \{5, 10\}$. They are computed per physical trajectory before averaging. Arrival, endpoint, and sustained-window labels address three control questions; block-boundary and final-step labels are two choices of endpoint checks. The OR/end distinction is quantified in Appendix F.

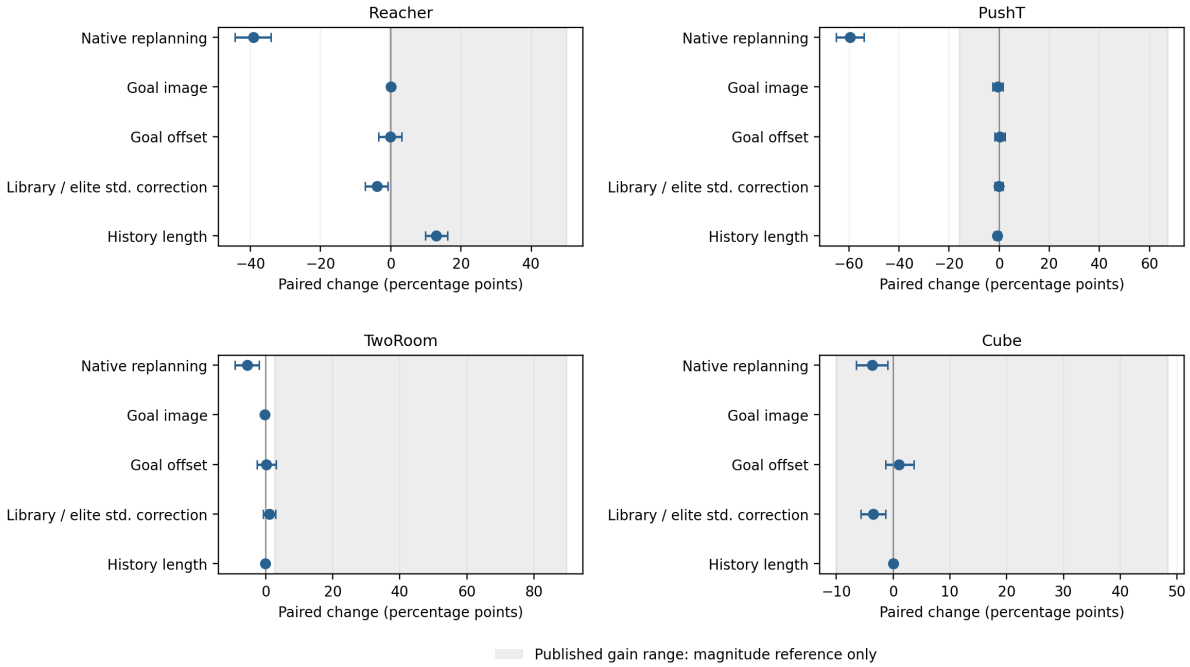


Fig. 2. Protocol-factor effects relative to the reference protocol. Error bars are case-bootstrap 95% intervals; the figure uses the same paired cases and streams for each factor. Gray bands are the minimum–maximum numeric gains in the selected literature inventory, included only as a magnitude reference; their bounds are the inventory extrema, not a pooled estimate.

3.6 Statistics and registration

For method m, checkpoint k, stream s, case i, and success definition d, let $Z_{ikmsd} \in [0,1]$ denote the binary label or mean window occupancy computed on one complete physical trajectory. Define $Y_{imsd} = K_m^{-1} \sum_{k=1}^{K_m} Z_{ikmsd} \in [0,1]$, where $K_m = 3$ for AD-WM and $K_m = 1$ otherwise. A nonlinear trajectory metric is calculated before checkpoint averaging; the averaged labels are not thresholded again. The estimand is the fixed-checkpoint, fixed-three-stream average over the case distribution, with the planning-stream mean formed inside each case:

$$\widehat{\Delta}_{m,d} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{1}{S} \sum_{s=1}^{S} Y_{imsd} - \frac{1}{S} \sum_{s=1}^{S} Y_{\mathrm{iref}sd} \right].$$

The 95% percentile interval resamples the same case indices for every method, stream, checkpoint, and success definition in each bootstrap replicate. AD-WM checkpoint seeds are averaged within case and stream before the contrast. This estimates the average effect for the fixed public checkpoints, fixed planner streams, and fixed protocol; it does not estimate retraining, a different stream set, a different retrieval bank, or another environment version. Contrasts use 5,000 case-bootstrap replicates. Degenerate cells also receive exact binomial intervals, while multi-stream or multi-checkpoint case means are never treated as independent binomial trials. The audit records source-specific reproduction, factor contrasts, arrival/endpoint/window labels, occupancy, rank correlations, and held-out prediction tests. The protocol sequence began with source reproduction, then native reset and held-out confirmation; after the scope was fixed, the held-out work retained the history, common-protocol, and

Table 5. Reproduction under each source protocol. Match = 1 (“yes”) when the published rate lies inside the measured 95% case-bootstrap interval, as specified in the study preregistration, Section 4 (original-protocol reproduction); no fixed ε tolerance was registered. Intervals shown are case-level 95% intervals. For the degenerate Fast-LeWM PushT cell, the displayed exact two-sided binomial interval supplements the original bootstrap interval [100, 100]; Match retains the registered bootstrap decision.

Method	Task	Published (%)	Measured (%, 95% CI)	Match
Fast-LeWM	TwoRoom	98.0	96.0 [90.0, 100.0]	yes
Fast-LeWM	Reacher	88.0	88.0 [78.0, 96.0]	yes
Fast-LeWM	PushT	96.0	100.0 [92.9, 100.0]	no
Fast-LeWM	Cube	80.0	78.0 [66.0, 88.0]	yes
AP	Cube	46.9	46.9 [38.3, 55.5]	yes
AP	PushT	69.5	70.3 [61.7, 78.1]	yes
AP	Reacher	75.8	97.7 [94.5, 100.0]	no
AP	TwoRoom	46.9	46.1 [37.5, 54.7]	yes
AD-WM	Cube	90.7	89.3 [81.3, 96.0]	yes
AD-WM	Reacher	83.3	81.3 [73.3, 88.7]	yes
AD-WM	TwoRoom	98.0	97.3 [92.7, 100.0]	yes
AD-WM	PushT	92.0	92.7 [88.0, 96.7]	yes

success-definition questions and included the CEM-iteration comparison. The complete chronology and scope decisions are in Appendix A.

The original preregistered any-step predictions retain confirmatory status under their original cases, configurations, and rules: the full saved-record audit finds zero old/new any-step outcome mismatches (Appendix F). The correction registration froze the fix and analysis plan before the valid formal rerun (Table 26). CEM-iteration comparisons, corrected endpoint effects, sustained-window results, and ranking changes are descriptive analyses, not a new held-out confirmation: those cases had already been seen.

4 Can the Published Numbers Be Reproduced?

Table 5 gives the source value, our measured value, its interval, and whether the published value lies within the preregistered bootstrap interval. Ten of 12 cells match. The Fast-LeWM PushT value is 100.0% versus a published 96.0%. AP Reacher is 97.7% versus 75.8% in the source record. Both use the released cases and query list. In the AP list, 30 source failures become local successes and two source successes become failures; the record does not identify a single cause.

5 Which Unreported Choices Move the Number?

5.1 Factor effects relative to published gains

Figure 2 places each paired factor effect on the four tasks; the corresponding intervals are in Table 6. The history change is the largest stable factor in Reacher, while native replanning lowers Reacher by -39.2 points and PushT by -59.5 points. The re-planning comparison includes the native every-five-step schedule and warm start; the experiment did not separate those implementation components.

5.2 History and software behavior

Three-frame history adds +12.7 points in Reacher any-step success (95% CI [+8.3, +17.3]), with changes of +0.0, -0.7, and +0.3 points in the other tasks. Corrected final-step success rises from 37.3% to 85.7%, a paired +48.3

Table 6. Same-stream paired factor effects across three fixed planner random-seed streams on the 200-case mixed set. The library-version and CEM elite-standard-deviation factors are numerically identical and appear once as a combined row; CEM iteration-count comparisons use a separate held-out set and appear in Appendix C. The Cube goal-image condition was not run after the recorded Cube goal-image incident.

Factor	Task	Difference in percentage points (95% CI)
History length	Reacher	+13.0 [+9.8, +16.2]
Library / CEM elite std. correction	Reacher	-4.0 [-7.3, -0.8]
Goal offset	Reacher	-0.2 [-3.5, +3.2]
Goal image	Reacher	+0.0 [+0.0, +0.0]
Native replanning	Reacher	-39.2 [-44.3, -34.2]
History length	PushT	-0.7 [-2.0, +0.7]
Library / CEM elite std. correction	PushT	-0.2 [-1.8, +1.5]
Goal offset	PushT	+0.3 [-1.7, +2.3]
Goal image	PushT	-0.5 [-2.7, +1.5]
Native replanning	PushT	-59.5 [-65.0, -54.0]
History length	TwoRoom	-0.2 [-0.8, +0.5]
Library / CEM elite std. correction	TwoRoom	+1.0 [-0.8, +2.8]
Goal offset	TwoRoom	+0.2 [-2.7, +3.0]
Goal image	TwoRoom	-0.3 [-0.8, +0.0]
Native replanning	TwoRoom	-5.5 [-9.2, -2.0]
History length	Cube	+0.0 [-0.5, +0.5]
Library / CEM elite std. correction	Cube	-3.5 [-5.7, -1.3]
Goal offset	Cube	+1.0 [-1.3, +3.7]
Goal image	Cube	not run (recorded Cube goal-image incident)
Native replanning	Cube	-3.7 [-6.5, -1.0]

[+42.0, +55.0]-point effect. The intervention provides observation and action history, and this study measures its effects without attributing them to a particular mechanism. One-frame any-step success is already 84.7%, leaving only +15.3 points below its ceiling.

The first query contains only the current frame. Across all 300 paired case-streams, the first 25 raw actions are bitwise identical for one- and three-frame LeWM, with maximum absolute difference zero; their first action divergence is at step 26. First-plan OR and END labels also agree. Thus the first plan is identical by construction and verification, and both gains arise in the second plan, at steps 26–50. Table 11 shows why these arrival and endpoint improvements should not be called sustained holding: three-frame LeWM's all-ten success is only 0.3%.

The library comparison changes the CEM elite standard-deviation correction. Its paired effects are about -4.0 points in Reacher and -3.5 points in Cube, while PushT and TwoRoom intervals include zero. The library change and elite-correction intervention are numerically identical and are merged into one row because this correction is the substantive version difference. CEM 10 versus 30 iterations changes the measured rate by at most about 6.7 points in these cells; the full iteration table and plot are in Appendix C. Goal offset and goal image changes are smaller than the history and replanning effects.

The Cube goal-image factor has no success-rate estimate: one isolated goal-render equality check failed before any model call or environment step, so that condition was excluded. The failure does not remove or alter the other Cube factor conditions, whose input images and trajectories passed their recorded checks.

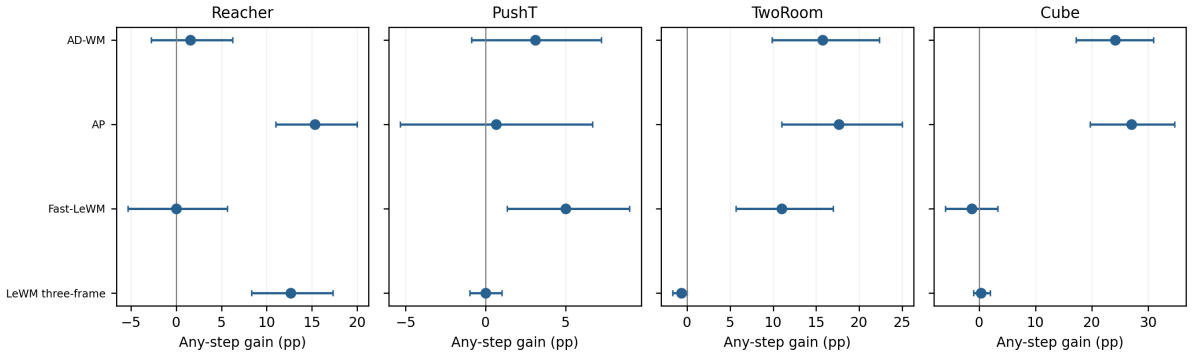


Fig. 3. Common-protocol paired gains under any-step success. Points are contrasts and bars are case-bootstrap 95% intervals.

Table 7. Preregistered prediction checks on the held-out set. Each row is a separately judged subcondition; estimates are paired percentage-point differences [95% case-bootstrap interval]. Reacher uses corrected trajectories, and every verdict is evaluated from the registered rule.

Prediction	Subcondition	Preregistered rule	Estimate [95% CI]	Result
P1	Reacher (3-frame − 1-frame)	$\widehat{\Delta} \geq +10$ pp and CI lower bound > 0	+12.7 [+8.3, +17.3]	supported
P1	PushT (3-frame − 1-frame)	$\lvert\widehat{\Delta}\rvert \leq 2$ pp (point-estimate rule)	+0.0 [-1.0, +1.0]	supported
P1	TwoRoom (3-frame − 1-frame)	$\lvert\widehat{\Delta}\rvert \leq 2$ pp (point-estimate rule)	-0.7 [-1.7, +0.0]	supported
P1	Cube (3-frame − 1-frame)	$\lvert\widehat{\Delta}\rvert \leq 2$ pp (point-estimate rule)	+0.3 [-1.0, +2.0]	supported
P2	Reacher: LeWM 3-frame − Fast-LeWM	$\widehat{\Delta} \geq 0$ pp and CI lower bound > -5 pp	+12.7 [+7.7, +17.7]	supported
P2	Reacher: LeWM 3-frame − AP	$\widehat{\Delta} < 0$ pp (point-estimate rule)	-2.7 [-4.7, -1.0]	supported
P2	Reacher: LeWM 3-frame − AD-WM	$\widehat{\Delta} \geq 0$ pp and CI lower bound > -5 pp	+11.1 [+7.3, +15.0]	supported
P3	TwoRoom: AP − LeWM 1-frame	CI lower bound > 0 pp (gain rule)	+17.7 [+11.0, +25.0]	supported
P3	TwoRoom: AD-WM − LeWM 1-frame	CI lower bound > 0 pp (gain rule)	+15.8 [+9.9, +22.3]	supported
P3	Cube: AP − LeWM 1-frame	CI lower bound > 0 pp (gain rule)	+27.0 [+19.7, +34.7]	supported
P3	Cube: AD-WM − LeWM 1-frame	CI lower bound > 0 pp (gain rule)	+24.1 [+17.2, +30.9]	supported

6 How Much of the Published Gain Survives a Common Protocol?

Figure 3 compares each method with LeWM one-frame on the same held-out cases and streams. AP and AD-WM gains on TwoRoom are +17.7 and +15.8 points; on Cube they are +27.0 and +24.1 points. Their intervals exclude zero. In Reacher, three-frame history itself is +12.7 points above one-frame LeWM; it is +12.7 points above Fast-LeWM and +11.1 points above AD-WM. Three-frame LeWM is 2.7 percentage points below AP under any-step success. PushT contrasts are within a few points.

Table 8. Arrival and endpoint labels on the held-out trajectories. Reacher uses corrected continuous physics; block and final labels use environment-step-end states.

Task	Configuration	Any-step (%)	Block (%)	Final (%)
Reacher	LeWM one-frame	84.7	63.7	37.3
Reacher	LeWM three-frame	97.3	92.7	85.7
Reacher	Fast-LeWM	84.7	68.7	42.7
Reacher	AP	100.0	96.7	66.0
Reacher	AD-WM	86.2	69.8	47.8
PushT	LeWM one-frame	90.3	89.3	68.3
PushT	LeWM three-frame	90.3	88.7	64.7
PushT	Fast-LeWM	95.3	95.0	92.3
PushT	AP	91.0	86.7	46.7
PushT	AD-WM	93.4	91.4	76.3
TwoRoom	LeWM one-frame	81.0	81.0	79.3
TwoRoom	LeWM three-frame	80.3	80.3	79.3
TwoRoom	Fast-LeWM	92.0	92.0	92.0
TwoRoom	AP	98.7	98.7	98.7
TwoRoom	AD-WM	96.8	96.8	96.7
Cube	LeWM one-frame	62.3	60.0	51.0
Cube	LeWM three-frame	62.7	60.3	52.3
Cube	Fast-LeWM	61.0	60.0	48.7
Cube	AP	89.3	88.0	75.3
Cube	AD-WM	86.4	85.7	81.1

All three preregistered predictions satisfy their registered rules on the held-out set (Table 7). The main, fresh, and held-out cohorts show the same direction for the history and TwoRoom/Cube contrasts. The AP retrieval bank differs between the two evaluation rounds because all earlier evaluation episodes were removed before held-out selection; the cohorts are therefore reported separately.

7 What Counts as Success?

7.1 Arrival and endpoint labels on one trajectory

Table 8 applies any-step, block-boundary, and final-step labels to identical 50-step traces. LeWM Reacher falls from 84.7% to 37.3% when the label changes from any-step to final-step. AP falls from 100.0% to 66.0%. The paired AP–LeWM difference is +15.3 [+11.0, +20.0] points under any-step and +28.7 [+20.7, +36.3] points under final-step; the same bootstrap gives a +13.3 [+5.7, +21.0]-point change in the AP–LeWM difference when the success definition changes. Figure 4 shows all four tasks and all five configurations.

7.2 Correcting our continuation adapter

Continuing a Reacher rollout past success requires disabling the task's success termination. Our earlier continuation adapter cleared only the outer `terminated` flag; dm_control had already scheduled a reset, so the arm was re-randomized on the next step. The released first-arrival evaluation never continues past success and is unaffected. In a recorded held-out trajectory, replay under the original semantics reproduces 50/50 recorded success labels. With success termination disabled, the within-step OR labels agree on all 6/6 labels through first success; 8 of the 44 later labels differ. This label agreement does not assert equality of the environment-step-end

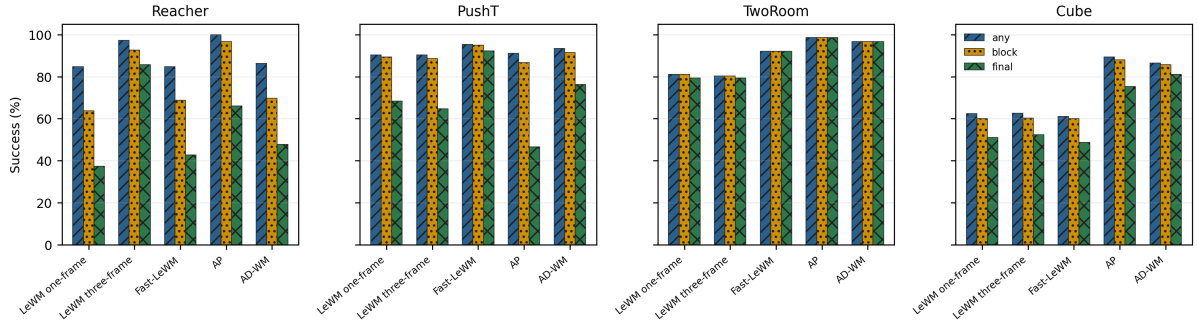


Fig. 4. Arrival (any-step) and endpoint (block-boundary and final-step) labels applied to the same held-out trajectories. Bar textures distinguish the three check-time choices in grayscale.

Table 9. Measurements under the earlier continuation adapter, kept for comparison. The arm was reset after each success, so the block and final columns describe reset episodes, not continuous rollouts.

	Earlier adapter (reset after success)			Corrected adapter		
Configuration	Any	Block	Final	Any	Block	Final
LeWM one-frame	84.7	26.3	9.3	84.7	63.7	37.3
LeWM three-frame	97.3	37.3	16.7	97.3	92.7	85.7
Fast-LeWM	84.7	30.3	10.0	84.7	68.7	42.7
AP	100.0	31.0	1.3	100.0	96.7	66.0
AD-WM	86.2	31.9	11.9	86.2	69.8	47.8

state at the first hit, because the corrected wrapper completes both microsteps. The corrected adapter retains the same predicate and dynamics while disabling this success termination.

The old/new comparison combines reset removal and the explicit END convention; the OR/END table in Appendix F isolates label choice on the corrected trajectories. Any-step rates are identical before and after correction in every configuration, as intended, and the full 2700-record audit confirms zero paired any-step outcome mismatches. Appendix F (Table 24) compares the AP–LeWM contrasts and rank correlations under the two adapters.

Table 21 traces the conditions behind this failure mode in public code. Success termination, an automatic reset, and continued sampling must coincide for post-success observations to cross an episode boundary; clearing an outer vector flag does not clear dm_control's internal reset state. Stopping at success or explicitly retaining episode boundaries avoids that interpretation; the table reports static paths, not a re-evaluation of published scores.

7.3 Ranking reversals

The same controller pair can reverse order across arrival, endpoint, and sustained-window labels. For three-frame LeWM minus AP in Reacher, any-step, final-step, and all-last-ten contrasts are -2.7 [-4.7, -1.0], +19.7 [+12.3, +27.3], and -10.3 [-14.3, -6.7] points; all three intervals exclude zero, with two opposite orderings. For AP minus one-frame LeWM in PushT, the corresponding contrasts are +0.7 [-5.3, +6.7], -21.7 [-31.3, -11.7], and +25.3 [+17.3, +33.3] points. Any-step success is unresolved, while final-step and sustained-window success establish opposite

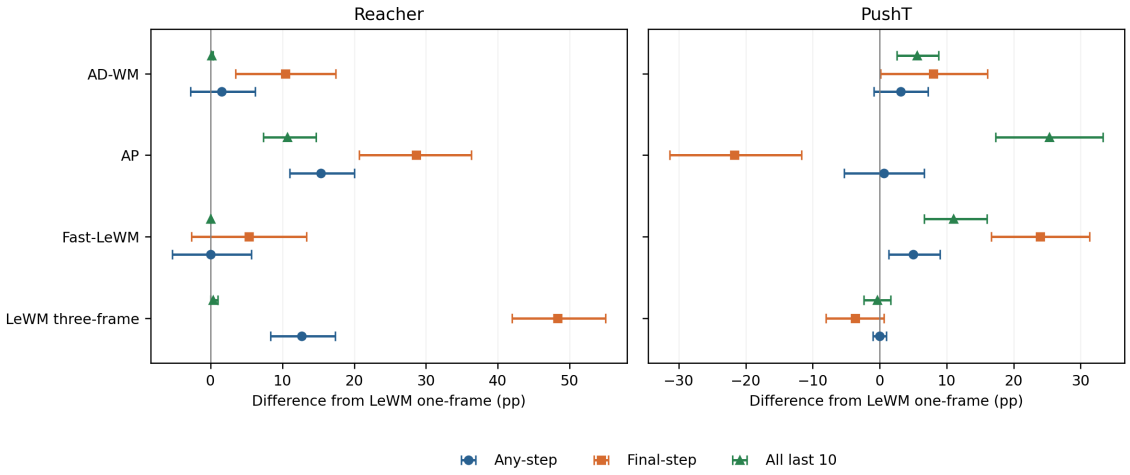


Fig. 5. Arrival (any-step), endpoint (final-step), and sustained-window (all-last-ten) labels can order the same controllers differently.

Table 10. Kendall rank correlations between success definitions, with corrected Reacher continuation.

Task	Definitions	τ_b
Reacher	any / block	0.95
Reacher	any / final	0.74
Reacher	block / final	0.80
PushT	any / block	0.53
PushT	any / final	0.53
PushT	block / final	1.00
TwoRoom	any / block	1.00
TwoRoom	any / final	0.95
TwoRoom	block / final	0.95
Cube	any / block	0.95
Cube	any / final	0.80
Cube	block / final	0.74

orderings. Figure 5 uses one-frame LeWM as the reference and shows arrival, endpoint, and sustained-window labels for every configuration. For PushT, the final-minus-any contrast change is -22.3 [-31.3, -13.3] points. TwoRoom's any-step and final-step rates and ordering are close.

Kendall correlations summarize the same result. Reacher's any-step versus final-step value is 0.74; PushT is 0.53; TwoRoom is 0.95; and Cube is 0.80. The block-boundary correlations are in Table 10. A benchmark therefore has a metric choice in addition to a checkpoint choice.

Sustained-window rankings. Table 11 adds the third question to first arrival and endpoint occupancy. In Reacher, three-frame LeWM leads AP at the final step, whereas AP leads in mean last-ten occupancy by +33.3 [+28.8, +37.8] points and in all-ten success by +10.3 [+6.7, +14.3]. AP also exceeds one-frame LeWM by +44.2 [+40.1, +48.3] and +10.7 [+7.3, +14.7] points, respectively. PushT AP and Fast-LeWM both exceed one-frame LeWM on the

Table 11. Endpoint and sustained-window occupancy on the same trajectories (percent). Means and nonlinear all-window labels are computed per trajectory before case-level checkpoint and stream averaging. These are descriptive corrected comparisons.

Task	Configuration	Final step	Mean last 10	All last 5	All last 10
Reacher	LeWM one-frame	37.3	13.3	1.3	0.0
Reacher	LeWM three-frame	85.7	24.1	6.3	0.3
Reacher	Fast-LeWM	42.7	14.6	3.3	0.0
Reacher	AP	66.0	57.4	27.7	10.7
Reacher	AD-WM	47.8	15.0	2.2	0.1
PushT	LeWM one-frame	68.3	25.4	14.3	2.0
PushT	LeWM three-frame	64.7	21.7	10.0	1.7
PushT	Fast-LeWM	92.3	44.2	32.3	13.0
PushT	AP	46.7	48.2	31.7	27.3
PushT	AD-WM	76.3	31.7	22.9	7.6

corresponding window measures; all paired intervals are in Appendix D. Sustained-window occupancy therefore supplies another ranking rather than a synonym for final-step success.

Table 20 checks threshold and window sensitivity: the Reacher final-versus-all-window reversal retains nonzero intervals at 0.05 and 0.08 rad for $K = 5, 10, 15, 20$; at 0.03 rad the all-window interval first touches zero at $K = 10$, with zero contrasts at $K = 15, 20$. PushT AP retains the reversal for all four windows at the recorded threshold, whereas Fast-LeWM leads under both labels; the AP mean-occupancy interval includes zero at $K = 5$.

7.4 Occupancy over time

Figure 6 reports the mean step-end predicate at each raw step, with checkpoint and stream averaging inside each case. Corrected one-frame LeWM Reacher occupancy is 22.0% at step 25 and 37.3% at step 50; its minimum during steps 26–49 is 2.3% at step 33. PushT, especially LeWM and AP, peaks near step 25, declines during steps 26–49, and rises again toward step 50; Fast-LeWM has peaks at both endpoints. TwoRoom mostly peaks at step 50, while Cube peaks at steps 25–26 and declines before a smaller endpoint rebound. The planned endpoints therefore coincide with many arrivals, while mid-plan exits create the gap between any-step and final-step labels. The exact peak and middle-minimum values are listed in Table 12. AP's Reacher curve has a five-step sawtooth, with peaks at re-decision boundaries; LeWM and AD-WM peak at steps 25 and 50. These curves describe success timing on the environment side; Vesessook and Yang (2026) study the planner's scoring side.

8 How Small a Difference Can One Evaluation Resolve?

A 50-case, one-stream Reacher evaluation has a typical paired interval half-width near 10.0 points. With 100 cases and one stream it is 7.5 points; with three streams it is 4.83 points. PushT gives 3.0, 2.5, and 1.67 points for the same three designs. These half-widths use any-step labels in the earlier Step-2 200-case cohort and are unaffected by the continuation correction. These "typical" values are medians of the paired bootstrap half-widths for the recorded 50-case subsets, with one or three streams averaged within each case; they are empirical resolution summaries, not minimum detectable effects. The variance of a paired difference is driven by the fraction of cases on which two methods disagree, so it is not determined by case count alone. The 100-case, three-stream design resolves broad gains such as those on Cube but cannot distinguish many one- or two-point claims. Figure 7 shows the comparison. The broader Step-0 candidate inventory (supplied in the materials) contains 99 numeric method–task gain claims on these four tasks. 56 (56.6%) have absolute magnitude at most 10 percentage points,

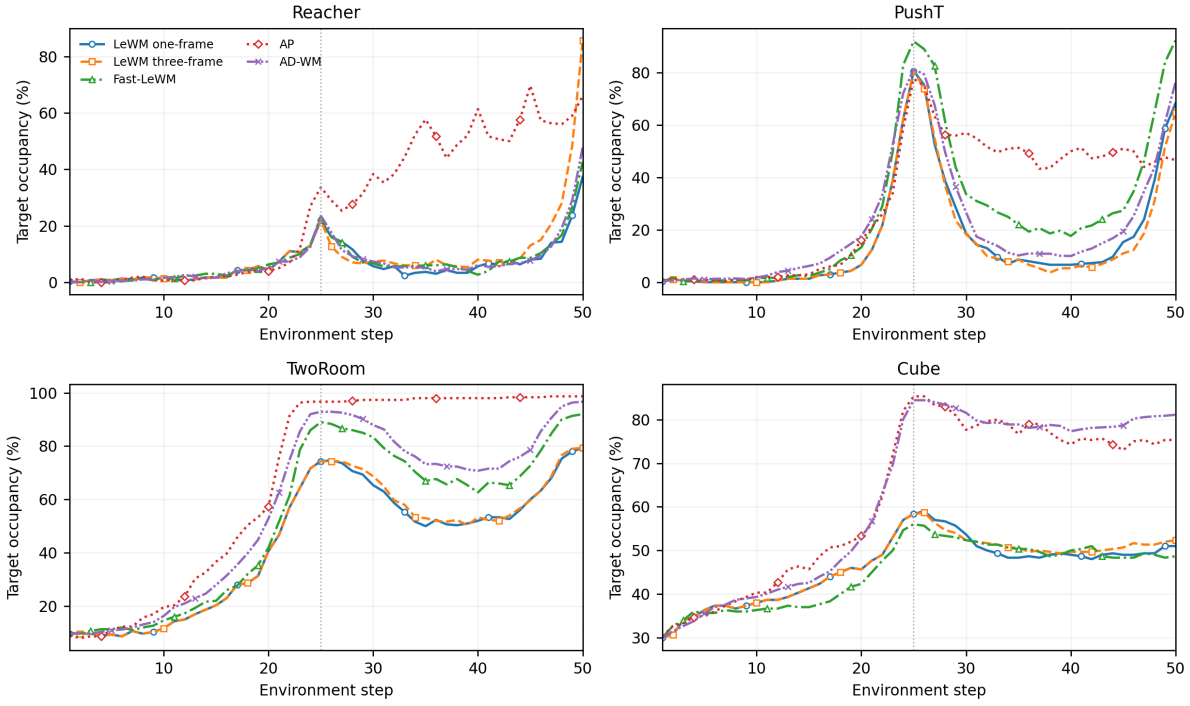


Fig. 6. Occupancy is the mean per-step predicate, averaging checkpoint and planning-stream indicators within each case; vertical guides mark steps 25 and 50. Each task has its own vertical scale; Reacher uses the corrected step-end predicate.

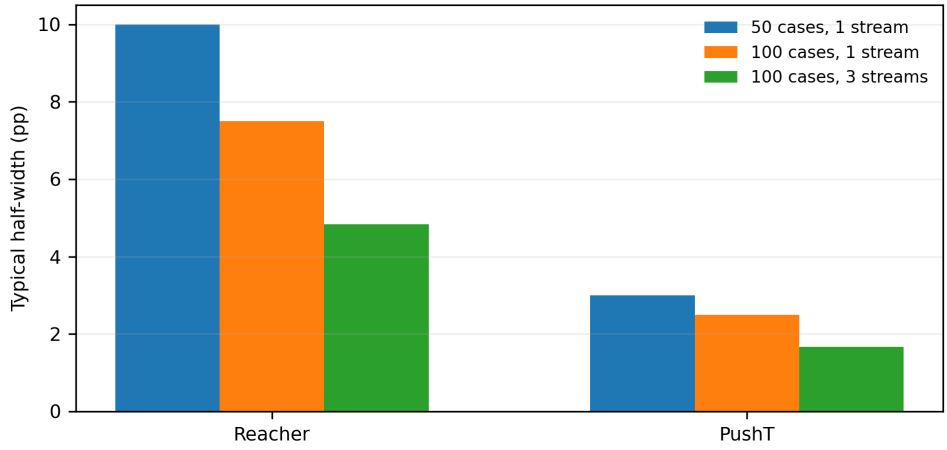


Fig. 7. Resolution improves with both cases and planning streams.

the approximate resolution of a 50-case one-stream Reacher comparison. Examples include Fast-LeWM gains

Table 12. Target occupancy at planned endpoints and within the intervening steps; Reacher uses corrected END labels.

Task	Configuration	Step 25	Step 50	Peak (step)	Min. 26–49 (step)
Reacher	LeWM one-frame	22.0	37.3	37.3 (50)	2.3 (33)
Reacher	LeWM three-frame	22.0	85.7	85.7 (50)	5.3 (39)
Reacher	Fast-LeWM	23.3	42.7	42.7 (50)	2.7 (40)
Reacher	AP	33.7	66.0	69.7 (45)	25.3 (27)
Reacher	AD-WM	23.6	47.8	47.8 (50)	3.9 (36)
PushT	LeWM one-frame	80.7	68.3	80.7 (25)	6.7 (38)
PushT	LeWM three-frame	80.7	64.7	80.7 (25)	3.7 (38)
PushT	Fast-LeWM	92.0	92.3	92.3 (50)	17.7 (40)
PushT	AP	77.7	46.7	77.7 (25)	43.3 (37)
PushT	AD-WM	81.4	76.3	81.4 (25)	10.0 (40)
TwoRoom	LeWM one-frame	74.3	79.3	79.3 (50)	50.0 (35)
TwoRoom	LeWM three-frame	74.3	79.3	79.3 (50)	50.3 (39)
TwoRoom	Fast-LeWM	89.0	92.0	92.0 (50)	62.7 (40)
TwoRoom	AP	96.7	98.7	98.7 (47)	96.7 (26)
TwoRoom	AD-WM	92.9	96.7	96.7 (50)	70.8 (40)
Cube	LeWM one-frame	58.3	51.0	59.0 (26)	48.0 (42)
Cube	LeWM three-frame	58.3	52.3	58.7 (26)	49.3 (39)
Cube	Fast-LeWM	56.0	48.7	56.0 (25)	48.3 (44)
Cube	AP	85.3	75.3	85.3 (25)	73.0 (45)
Cube	AD-WM	84.4	81.1	84.4 (25)	77.3 (40)

of +2 points on Reacher and 0 on PushT, and the LEAP PushT gain of +2 points. These are inventory values, reported without reinterpreting their protocols or assessing significance in any source paper.

9 Discussion

9.1 What each ruler measures

Any-step success asks whether the controller can bring the system through the target before the budget ends. Final-step success asks whether the system is in the target at a specified later time. Block-boundary success asks the same question at the five-step action-block boundaries. Sustained-window occupancy asks how much or whether all of a final window is spent in the target; it distinguishes endpoint arrival from remaining there throughout the window. These are different control questions. Methods in this benchmark are commonly tuned and reported with the any-step convention, so a paper should report final-step success alongside the conventional number. The step-wise reward in the original dm_control Reacher task illustrates the same distinction: transient proximity can contribute reward even when the terminal configuration differs.

9.2 A reporting checklist

The minimum fields and observed scales are in Table 13. A paper should state the task threshold, all check times, early termination, history construction, CEM candidates/elites/iterations, horizon, replanning interval, software commit and dependency versions, case selection, planning streams, training seeds, aggregation order, and uncertainty interval. An occupancy curve or first-hit diagnostic allows readers to interpret an any-step value alongside the final state.

Table 13. Minimum reporting fields and measured scales. The success-timing and history rows use corrected held-out results; factor and noise rows retain their original sealed cohorts.

Item	Why it matters / scope	Observed scale
Success test and check times	Any-step minus final-step rates across the 20 held-out task/configuration cells	0.0 to 47.3 pp
History construction	Held-out Reacher three-frame minus one-frame	+12.7 pp (any-step), +48.3 pp (final-step)
Termination and reset semantics	Corrected minus earlier-adapter Reacher block/final rates; includes both the continuity and END-label corrections	+28.0 to +69.0 pp
Microsteps versus step end	END minus OR block/final rates on identical corrected Reacher trajectories	-6.3 to -1.0 pp
Planner budget and replanning	Largest absolute native-replanning effect in the sealed mixed-case factor analysis	59.5 pp
Cases and planning streams	Typical paired half-width for 100 cases and three streams, Reacher and PushT	1.7 to 4.8 pp
Software and aggregation	Absolute elite-standard-deviation/version effects in the sealed factor analysis; report checkpoint averaging order	0.2 to 4.0 pp

9.3 Limitations

The empirical comparisons cover one world-model family, four simulated tasks, three public method families, and a finite set of planning streams. The static continuation audit additionally traces the public code paths in Table 21; that source-code coverage is distinct from empirical validation on those systems. The held-out exclusion concerns this study's own case and retrieval choices; public pretraining exposure is unknown. The AP retrieval bank differs between the two evaluation rounds. Main-result thresholds are inherited from the released environments; Appendix D varies the Reacher threshold, while missing PushT physical states prevent its threshold re-scoring. The causes of leaving a target and of the native replanning decrease are not separated. AP Reacher reproduction was 97.7% versus the source's 75.8%; this unexplained first-arrival discrepancy is unrelated to the continuation reset. AP conclusions here concern the implemented and recorded configuration. Results may differ for other world-model families, planners, or physical robots.

10 Conclusion

Evaluation choices determine what a latent-planning benchmark score means. In these four tasks, history length materially changes Reacher's any-step success, common-protocol gains on TwoRoom and Cube remain large, and first-arrival, endpoint and sustained-window labels can change the ordering of the same controllers. The Reacher audit also shows why a post-success continuation adapter must disable the task's success termination. Reporting the success timing, history, planner budget, case selection, and planning streams with every percentage makes such comparisons auditable and scientifically interpretable.

CRediT authorship contribution statement

Ruiqi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – initial manuscript, Writing – review and editing, Visualization, Project administration.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Audit records and code are archived at Zenodo, DOI 10.5281/zenodo.23350373, with records under CC BY 4.0 and audit code under MIT. The code mirror is https://github.com/2GyounnnG/lewm-protocol-audit. Public checkpoints and input datasets are identified by source URLs and SHA-256 hashes and are not redistributed.

Generative AI statement

During the preparation of this manuscript, the author used Claude (Anthropic) and OpenAI Codex (OpenAI) to assist with the design of analyses, writing and debugging experiment, analysis, and plotting code, preparing and editing text, and checking the manuscript against journal guidelines. The author reviewed and edited the manuscript and takes full responsibility for its content.

A Protocol timeline and scope

The evaluation was organized as a sequence of source reproduction, protocol-factor checks, stream-matched post hoc analyses, and held-out confirmation. PushT exact-start override was withdrawn and 21 affected conditions were rerun with native reset; 572 earlier partial records were excluded. The held-out definition excludes the study's prior cases and retrieval bank, plus episodes from the author's earlier retraining experiments on this benchmark, whose results are not reported here. Before held-out results existed, the replanning-mechanism grid and AP/AD-WM three-frame conditions were removed and a ten-versus-thirty CEM iteration comparison was added. Adapter-fidelity checks were carried over only for unchanged adapters; the common evaluation environment was rebuilt. Table 14 gives UTC times and document hashes.

Table 14. Reproducible protocol timeline. Times are UTC and hashes are the first 12 characters of the SHA–256 for the listed record. The complete hashes and all deviations are in the machine-readable audit file.

Event	UTC time or interval	SHA–256
V1 preregistration	2026-10-11 21:05:48Z	ab599de6bf99
V1.1 preregistration	2026-10-11 22:23:25Z	1e2c0131650b
V1.2 preregistration	2026-10-12 01:51:38Z	09235d12a429
V2 follow-up preregistration	2026-10-12 15:29:30Z	924ed56ca540
Author scope revision before held-out results	2026-10-12 16:21:37.665763Z	3b91dd0e5ddf
Held-out scope clarification	2026-10-12 17:17:12.557896Z	e492dda56dc3
Held-out case manifest sealed	2026-10-12 18:12:15.790699Z	98cb9adf2093
Formal run launch receipt	2026-10-12 18:58:57.462596Z	d9bb38d33a2b
Formal run / final return	2026-10-12 18:59:51.968689Z to 2026-10-12 20:25:46.842875Z	27ba4f422893

B Complete factor and method results

Table 8 gives the full held-out rate table used in the main figures. The accompanying machine-readable CSVs retain successful-trajectory counts, case-majority rates, and interval endpoints for every task, configuration, and

definition. Factor-level rows appear in Table 6; source protocol rows appear in Table 2. The AP–LeWM intervals

Table 15. Paired AP minus one-frame LeWM contrasts from the saved formal trajectories, using the frozen case bootstrap with 5,000 replicates and seed 20261011 for all tasks and labels.

Task	Any-step difference (95% CI)	Final-step difference (95% CI)	Change, final − any (95% CI)
Reacher	+15.3 [+11.0, +20.0]	+28.7 [+20.7, +36.3]	+13.3 [+5.7, +21.0]
PushT	+0.7 [-5.3, +6.7]	-21.7 [-31.3, -11.7]	-22.3 [-31.3, -13.3]
TwoRoom	+17.7 [+11.0, +25.0]	+19.3 [+12.3, +26.7]	+1.7 [+0.3, +3.7]
Cube	+27.0 [+19.7, +34.7]	+24.3 [+17.3, +31.7]	-2.7 [-8.3, +3.0]

in Table 15 use one case-bootstrap index vector for both labels and for the reported change. Reacher rows use corrected continuous trajectories and the sealed bootstrap seed; the original paired analyses for the other tasks are retained. The AD-WM rows retain each public checkpoint and each of the three planner streams rather than treating them as independent cases. The machine-readable audit contains the same fields for the other evaluated methods; Reacher rows use the corrected continuation semantics.

Table 16. AD-WM public checkpoint IDs (3072, 4096, 6144) and planner-stream results on 100 held-out cases per row. Reacher is replaced by the corrected run; all other task rows retain the sealed Step-4 results.

Task	Public checkpoint ID	Planner stream	Cases	Any-step (%)	Block (%)	Final (%)
Reacher	3072	planning stream 1	100	84.0	72.0	47.0
Reacher	3072	planning stream 2	100	83.0	72.0	52.0
Reacher	3072	planning stream 3	100	89.0	71.0	46.0
Reacher	4096	planning stream 1	100	85.0	67.0	50.0
Reacher	4096	planning stream 2	100	89.0	71.0	47.0
Reacher	4096	planning stream 3	100	92.0	72.0	45.0
Reacher	6144	planning stream 1	100	89.0	70.0	47.0
Reacher	6144	planning stream 2	100	84.0	67.0	51.0
Reacher	6144	planning stream 3	100	81.0	66.0	45.0
PushT	3072	planning stream 1	100	94.0	93.0	81.0
PushT	3072	planning stream 2	100	96.0	92.0	84.0
PushT	3072	planning stream 3	100	96.0	96.0	86.0
PushT	4096	planning stream 1	100	92.0	89.0	70.0
PushT	4096	planning stream 2	100	92.0	91.0	71.0
PushT	4096	planning stream 3	100	94.0	92.0	70.0
PushT	6144	planning stream 1	100	90.0	88.0	76.0
PushT	6144	planning stream 2	100	94.0	92.0	73.0
PushT	6144	planning stream 3	100	93.0	90.0	76.0
TwoRoom	3072	planning stream 1	100	98.0	98.0	98.0
TwoRoom	3072	planning stream 2	100	98.0	98.0	98.0
TwoRoom	3072	planning stream 3	100	97.0	97.0	97.0
TwoRoom	4096	planning stream 1	100	97.0	97.0	97.0
TwoRoom	4096	planning stream 2	100	97.0	97.0	97.0
TwoRoom	4096	planning stream 3	100	96.0	96.0	96.0
TwoRoom	6144	planning stream 1	100	95.0	95.0	95.0
TwoRoom	6144	planning stream 2	100	97.0	97.0	96.0
TwoRoom	6144	planning stream 3	100	96.0	96.0	96.0
Cube	3072	planning stream 1	100	85.0	84.0	82.0
Cube	3072	planning stream 2	100	86.0	86.0	80.0
Cube	3072	planning stream 3	100	90.0	90.0	85.0
Cube	4096	planning stream 1	100	90.0	89.0	86.0
Cube	4096	planning stream 2	100	84.0	84.0	78.0
Cube	4096	planning stream 3	100	85.0	84.0	80.0
Cube	6144	planning stream 1	100	87.0	87.0	80.0
Cube	6144	planning stream 2	100	83.0	81.0	77.0
Cube	6144	planning stream 3	100	88.0	86.0	82.0

C CEM iteration comparison

Figure 8 reports its paired differences for any-step, block-boundary, and final-step labels; CEM settings, history, cases, and streams are held fixed within each point.

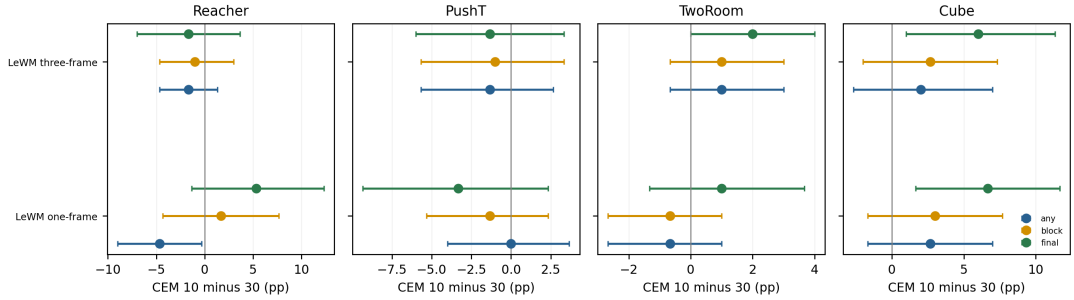


Fig. 8. CEM iteration comparison.

Table 17. Ten minus thirty CEM iterations, in percentage points with paired 95% case-bootstrap intervals. Reacher uses corrected continuous trajectories.

Task	History	Any-step	Block-boundary	Final-step
Reacher	1	-4.7 [-9.0, -0.3]	+1.7 [-4.3, +7.7]	+5.3 [-1.3, +12.3]
Reacher	3	-1.7 [-4.7, +1.3]	-1.0 [-4.7, +3.0]	-1.7 [-7.0, +3.7]
PushT	1	0.0 [-4.0, +3.7]	-1.3 [-5.3, +2.3]	-3.3 [-9.3, +2.3]
PushT	3	-1.3 [-5.7, +2.7]	-1.0 [-5.7, +3.3]	-1.3 [-6.0, +3.3]
TwoRoom	1	-0.7 [-2.7, +1.0]	-0.7 [-2.7, +1.0]	+1.0 [-1.3, +3.7]
TwoRoom	3	+1.0 [-0.7, +3.0]	+1.0 [-0.7, +3.0]	+2.0 [0.0, +4.0]
Cube	1	+2.7 [-1.7, +7.0]	+3.0 [-1.7, +7.7]	+6.7 [+1.7, +11.7]
Cube	3	+2.0 [-2.7, +7.0]	+2.7 [-2.0, +7.3]	+6.0 [+1.0, +11.3]

D Supplementary terminal-window metrics

The final-K occupancy, all-K success, and post-first-hit exit metrics are summaries of the same continuous trajectories. These are additional trajectory-level summaries computed with the unchanged task predicate; they were not part of the preregistered predictions. Metrics are calculated on each checkpoint trajectory before averaging streams and training seeds. A first hit at the last step has no subsequent observation and is excluded from the conditional exit denominator. The tables for all tasks use this same aggregation rule.

Table 18. Terminal-window summaries computed per physical trajectory before checkpoint averaging. Occupancy is the mean END indicator; all-K requires every END label in the window. Exit-after-hit is the proportion of eligible trajectories with at least one later END failure. Outside-after-hit is their mean outside fraction strictly after first ANY hit.

Task	Configuration	Occ. $K = 5$	Occ. $K = 10$	All $K = 5$	All $K = 10$	Exit after hit	Outside after hit
Reacher	LeWM one-frame	19.5	13.3	1.3	0.0	90.1	76.3
Reacher	LeWM three-frame	39.5	24.1	6.3	0.3	88.6	69.2
Reacher	Fast-LeWM	21.9	14.6	3.3	0.0	93.2	78.1
Reacher	AP	58.9	57.4	27.7	10.7	98.7	50.9
Reacher	AD-WM	23.6	15.0	2.2	0.1	90.9	74.9
PushT	LeWM one-frame	41.5	25.4	14.3	2.0	98.1	69.8
PushT	LeWM three-frame	35.7	21.7	10.0	1.7	98.9	72.7
PushT	Fast-LeWM	64.5	44.2	32.3	13.0	92.9	53.5
PushT	AP	46.9	48.2	31.7	27.3	81.3	42.1
PushT	AD-WM	48.2	31.7	22.9	7.6	94.7	61.9
TwoRoom	LeWM one-frame	72.8	63.9	61.3	47.3	53.5	19.2
TwoRoom	LeWM three-frame	73.4	64.2	62.7	48.3	54.8	18.0
TwoRoom	Fast-LeWM	87.3	77.5	76.7	60.7	50.5	15.0
TwoRoom	AP	98.6	98.4	98.3	98.0	15.9	1.4
TwoRoom	AD-WM	92.8	83.6	85.0	65.7	54.5	12.7
Cube	LeWM one-frame	49.9	49.4	47.7	45.7	32.8	16.7
Cube	LeWM three-frame	51.7	50.9	50.0	47.0	31.4	15.9
Cube	Fast-LeWM	48.7	49.0	46.3	46.0	27.3	13.6
Cube	AP	75.1	74.9	69.7	66.3	36.6	12.4
Cube	AD-WM	80.7	79.5	79.3	74.7	18.8	6.8

All values are percentages. Eligibility requires a first hit before step 50; a first hit only at step 50 has no observable later step and is excluded from both post-hit denominators. – denotes an empty eligible set. AD-WM metrics are computed separately for each checkpoint trajectory before averaging; eligible trajectory counts are supplied with the table data.

Table 19. Sustained-window contrasts (percentage points, paired 95% intervals). The frozen case-bootstrap function uses 5,000 replicates, seed 20261011, and the same index vectors for every definition.

Task	Contrast	Metric	Difference [95% CI]
Reacher	AP − LeWM three-frame	Mean last 10	+33.3 [+28.8, +37.8]
Reacher	AP − LeWM three-frame	All last 10	+10.3 [+6.7, +14.3]
Reacher	AP − LeWM one-frame	Mean last 10	+44.2 [+40.1, +48.3]
PushT	AP − LeWM one-frame	Mean last 10	+22.7 [+14.6, +30.6]
PushT	Fast-LeWM − LeWM one-frame	Mean last 10	+18.8 [+13.8, +24.0]
Reacher	LeWM three-frame − LeWM one-frame	All last 10	+0.3 [+0.0, +1.0]
Reacher	Fast-LeWM − LeWM one-frame	All last 10	+0.0 [+0.0, +0.0]
Reacher	AP − LeWM one-frame	All last 10	+10.7 [+7.3, +14.7]
Reacher	AD-WM − LeWM one-frame	All last 10	+0.1 [+0.0, +0.3]
PushT	LeWM three-frame − LeWM one-frame	All last 10	-0.3 [-2.3, +1.7]
PushT	Fast-LeWM − LeWM one-frame	All last 10	+11.0 [+6.7, +16.0]
PushT	AP − LeWM one-frame	All last 10	+25.3 [+17.3, +33.3]
PushT	AD-WM − LeWM one-frame	All last 10	+5.6 [+2.6, +8.8]

Table 20. Threshold and window sensitivity. Paired differences in percentage points with case-bootstrap 95% intervals (5,000 draws, common indices). Reacher uses recomputed END predicates; PushT retains its saved predicate because physical states were not saved. F denotes final-step success; columns 5–20 give all-K or mean-K occupancy.

Comparison / threshold	Metric	$K = 5$	$K = 10$	$K = 15$	$K = 20$
Reacher h3–AP / .03	F		+24.67 [15.00, 34.33]		
	All	-4.33 [-7.33, -1.67]	-0.67 [-1.67, 0.00]	+0.00 [0.00, 0.00]	+0.00 [0.00, 0.00]
	Mean	-5.87 [-10.07, -1.47]	-15.43 [-18.83, -11.90]	-17.33 [-20.27, -14.33]	-18.00 [-20.65, -15.28]
Reacher h3–AP / .05	F		+19.67 [12.33, 27.33]		
	All	-21.33 [-27.67, -15.33]	-10.33 [-14.33, -6.67]	-5.67 [-8.67, -3.00]	-2.33 [-4.00, -1.00]
	Mean	-19.40 [-24.73, -14.00]	-33.33 [-37.80, -28.80]	-37.20 [-41.36, -33.07]	-37.70 [-41.40, -33.95]
Reacher h3–AP / .08	F		+8.00 [4.00, 12.00]		
	All	-43.67 [-51.00, -36.33]	-51.67 [-58.01, -45.00]	-39.00 [-45.33, -32.67]	-26.00 [-31.67, -20.33]
	Mean	-23.93 [-28.53, -19.40]	-43.13 [-47.40, -38.77]	-49.91 [-54.13, -45.60]	-51.25 [-55.30, -47.05]
PushT AP–h1 / 20	F		-21.67 [-31.33, -11.67]		
	All	+17.33 [8.33, 26.67]	+25.33 [17.33, 33.33]	+23.67 [16.33, 31.33]	+21.33 [14.67, 28.67]
	Mean	+5.40 [-3.00, 13.60]	+22.73 [14.63, 30.57]	+28.36 [20.60, 36.09]	+31.57 [23.93, 39.28]
PushT Fast–h1 / 20	F		+24.00 [16.67, 31.33]		
	All	+18.00 [11.33, 25.00]	+11.00 [6.67, 16.00]	+7.00 [3.67, 10.67]	+7.00 [3.67, 10.67]
	Mean	+23.00 [16.93, 29.20]	+18.80 [13.83, 23.97]	+16.58 [11.93, 21.38]	+16.50 [11.73, 21.47]

E JAIR reproducibility checklist

The following items reproduce the JAIR author-template checklist; answers describe the current manuscript and deposited material package.

All articles:

(1) All claims investigated in this work are clearly stated. **Yes**. Sections 1, 3 and the preregistered prediction table.
(2) Clear explanations are given how the work reported substantiates the claims. **Yes**. Sections 4–8 and the paired result tables.
(3) Limitations or technical assumptions are stated clearly and explicitly. **Yes**. Sections 3 and 9.
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **NA**. No new AI method is introduced; the adapter mapping and evaluation logic are documented.
(5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. **Yes**. Sections 3 and 5, and Appendix A.

Articles containing theoretical contributions:

Does this paper make theoretical contributions? **No.** The conditional items below are not applicable.

(1) All assumptions and restrictions are stated clearly and formally. **NA**. No theoretical contribution.
(2) All novel claims are stated formally (e.g., in theorem statements). **NA**. No theoretical contribution.
(3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise (e.g., that expected from a PhD candidate in the same area of AI). **NA**. No theoretical contribution.
(4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. **NA**. No theoretical contribution.
(5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism (i.e., use that does not enhance clarity or precision) is avoided. **NA**. No theoretical contribution.
(6) Appropriate citations are given for all non-trivial theoretical tools and techniques. **NA**. No theoretical contribution.

Articles reporting on computational experiments:

Does this paper include computational experiments? **Yes.**

(1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for source code readability and documentation as well as for long-term accessibility. **Yes**. MIT audit code and reproduction instructions are archived at Zenodo, DOI 10.5281/zenodo.23350373, and mirrored on GitHub.
(2) The source code comes with a license that allows free usage for reproducibility purposes. **Yes**. The audit code is licensed under MIT; original third-party notices are retained.
(3) The source code comes with a license that allows free usage for research purposes in general. **Yes**. The audit code is licensed under MIT; original third-party notices are retained.
(4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility. **Yes**. Unaggregated audit records are archived at Zenodo, DOI 10.5281/zenodo.23350373.

(5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. **Yes**. Derived audit records are licensed under CC BY 4.0; public input datasets and weights are not redistributed.
(6) The unaggregated data comes with a license that allows free usage for research purposes in general. **Yes**. Derived audit records are licensed under CC BY 4.0; public input datasets and weights are not redistributed.
(7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. **Yes**. The frozen case, planner-stream, checkpoint and bootstrap seeds are supplied.
(8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. **Partially**. GPU and software records are included; a complete cache/RAM inventory is not reported for every earlier machine.
(9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. **Yes**. Section 3.5 and Appendices D and F.
(10) The number of algorithm runs used to compute each result is reported. **Yes**. Cohort and checkpoint/stream tables report the run grid.
(11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. **Yes**. Excluded conditions, the revoked premature launch and all retained failures are documented.
(12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. **Yes**. Paired confidence intervals, per-stream/checkpoint tables and occupancy curves are reported.
(13) All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. **Yes**. The method mapping, protocol-factor table and frozen condition specifications supply settings.
(14) The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. **Yes**. The protocol-factor grid and scope changes are documented; no checkpoint training or retraining was performed.
(15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. **Yes**. Prespecified paired-bootstrap criteria assess the confirmatory contrasts; the scope of the remaining contrasts is specified.

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? **Yes.**

(1) All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility with a license that allows free usage for research purposes. **Yes**. Derived records and case manifests are archived at Zenodo, DOI 10.5281/zenodo.23350373, under CC BY 4.0.
(2) The newly introduced data set comes with a license that allows free usage for reproducibility purposes. **Yes**. The derived records are licensed under CC BY 4.0.
(3) The newly introduced data set comes with a license that allows free usage for research purposes in general. **Yes**. The derived records are licensed under CC BY 4.0.
(4) All data sets drawn from the literature or other public sources (potentially including authors' own previously published work) are accompanied by appropriate citations. **Yes**. Section 2 and the source inventory cite the public benchmark and method sources.

(5) All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. **Yes**. Source URLs and asset hashes identify the publicly distributed input data.
(6) All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. **Yes**. The cohort, case selection, labels and derivation of audit records are documented.
(7) All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. **Yes**. Section 3 and the case/retrieval manifests describe preprocessing and episode-level exclusions.

Explanations on any of the answers above (optional):

The records are archived at Zenodo, DOI 10.5281/zenodo.23350373.

F Reacher runtime verification and continuation code audit

The TECH replay covers 108 trajectories, not the full formal grid. It instantiates the uninstrumented native stable-worldmodel Reacher wrapper, bypasses the recording and continuation wrapper, disables only success termination, and sends each saved action through two native dm_control steps. Comparisons include `qpos`, `qvel`, physical time and `target_qpos`, before and after every microstep. The maximum discrepancy is zero with absolute tolerance 10^{-12}. Predicates are separately recomputed from saved joint and target states. This is an independent replay path through the same simulator and dynamics, not an independent physics implementation.

All 2700 formal records additionally undergo a saved-log audit: exactly one initial reset, no internal or pending resets, monotone time with 0.02 s microsteps (0.04 s environment steps), constant targets, state-chain consistency, and recomputed predicates and OR/END labels. Across 270,000 microsteps every check has zero mismatches. The 82,691 actions before first success (85,094 including its action) are bitwise identical to the sealed original actions. Old/new any-step outcomes have zero mismatches per case, stream and checkpoint. These full-grid checks do not rerun physics. Patch diffs, the environment freeze and a minimal saved-log verification entry point are archived at Zenodo, DOI 10.5281/zenodo.23350373, and mirrored on GitHub.

Table 21. Static continuation audit at pinned source versions. T: success can terminate; R: reset behavior; E: evaluation flow; H: boundary handling. Commits, exact paths, line ranges, and dependency qualifications for all 22 audit entries are supplied in the materials. The rows describe code paths, not measured errors in source-paper scores.

Source/path	T	R	E	H
stable-worldmodel 63988116 and main: Reacher, PushT, TwoRoom, Cube	Yes; Cube depends on ManipSpace	World auto resets; Reacher also has inner reset	Dataset evaluation defaults to wait	Completed slots frozen
Other dm_control wrappers	No shared success hook	Internal LAST resets next step	World sees outer flags	Not all internal LAST flags forwarded
stable-worldmodel 0.0.6	Reacher qpos match terminates	Inner reset independent of vector bit	Fixed budget	Clears only vector autoreset bit
LeWM current eval	Dependency-defined	Dependency-defined	Calls World dataset evaluate	Default wait with audited modern World
AP original entry	Reacher explicit predicate	Underlying dm_control reset possible	Stops on success or done	Checks every primitive action
Fast-LeWM / AD-WM	Pinned 0.0.6 tasks	Inner reset remains possible	Fixed-budget loops	Clear only outer vector bit
DINO-WM PushT path	Success does not set done	No reset in traced rollout	Executes action sequence	No success-termination premise
TD-MPC2 dm_control	Task-dependent	Internal LAST resets next step	Outer timeout	Does not forward inner LAST
DreamerV3 dm_control	Native LAST/discount	Reset request or prior done	Episode driver	Explicit first/last/terminal labels
Meta-World	Optional success termination	Benchmark vector SAME_STEP	Evaluation enables success termination	Uses dones and final_info
OGBench ManipSpace	Default terminate-at-goal	No terminal-triggered pending reset in traced single-env step	Evaluation stops on done	Explicit next-episode reset
Gymnasium vector envs	Subenvironment-defined	Default NEXT_STEP	Caller-defined	Mode and final-observation handling required
PPWM	Author public code not found in the documented search.			

Table 22. Coverage of final-code TECH replay and the full saved-formal-log audit. Every row uses three planning streams. TECH uses four case identities per row; formal uses 100. No new physical rollout was run for this audit.

Method/checkpoint	History; CEM iterations	TECH replay	Formal log audit
LeWM	1; 30	12	300
LeWM	3; 30	12	300
Fast-LeWM	1; 30	12	300
AP	1; 30	12	300
AD-WM 3072	1; 30	12	300
AD-WM 4096	1; 30	12	300
AD-WM 6144	1; 30	12	300
LeWM	1; 10	12	300
LeWM	3; 10	12	300
Total		108	2700

Table 23. Reacher correction checks on the final recording code: 108 TECH trajectories, plus four-environment continuation and one original held-out trajectory replay. The two step labels are within-step any-success and step-end success.

Check	Numerical evidence	Result
T1: action prefix	3,397 actions before first success and 3,497 including its action are bitwise identical to the sealed original references (108/108 trajectories).	PASS
T2: reset counts	108 wrapper and 108 dm-control resets: one initial reset per trajectory; 0 internal resets.	PASS
T3: physical replay	5,400 environment steps / 10,800 microsteps; monotone time, 0.02 s per microstep, fixed target. Independent CPU replay maximum state error 0 (absolute tolerance 10^{-12}).	PASS
T4: success labels	10,800 predicates, 0 mismatches. Both step-label sequences and any/block/final outputs agree with their definitions; the two step labels differ at 78 environment steps.	PASS
Four-environment check	Original Reacher resets after success; corrected Reacher, PushT and TwoRoom continue with the recorded state continuity checks. Cube continues with increasing physical time.	PASS
Held-out replay	Original semantics reproduce 50/50 labels (100%). Corrected semantics agree through first success (6/6); thereafter 36/44 agree and 8 differ.	PASS

Table 24. AP–LeWM contrasts and five-configuration rank correlations under the earlier and corrected continuation adapters. Earlier block and final values describe episodes reset after success; corrected values describe continuous rollouts. Contrast entries are percentage points [95% paired case-bootstrap interval].

Statistic	Earlier adapter	Corrected adapter
AP minus LeWM: any	+15.3 [+11.0, +20.0]	+15.3 [+11.0, +20.0]
AP minus LeWM: final	-8.0 [-11.7, -4.7]	+28.7 [+20.7, +36.3]
AP minus LeWM: final minus any change	-23.3 [-28.7, -18.0]	+13.3 [+5.7, +21.0]
Kendall τ_b: any/block	0.53	0.95
Kendall τ_b: any/final	0.11	0.74
Kendall τ_b: block/final	0.60	0.80

Table 25. Within corrected Reacher trajectories, labels from either microstep (OR) versus labels at the second microstep (END). Any-step success uses OR; block and final success use END. Differences are END minus OR in percentage points.

Configuration	Metric	OR (%)	END (%)	Difference [95% CI]
LeWM one-frame	any	84.7	83.0	-1.7 [-3.3, -0.3]
LeWM one-frame	block	69.7	63.7	-6.0 [-9.0, -3.3]
LeWM one-frame	final	40.7	37.3	-3.3 [-5.7, -1.3]
LeWM three-frame	any	97.3	97.0	-0.3 [-1.0, +0.0]
LeWM three-frame	block	96.3	92.7	-3.7 [-7.0, -1.0]
LeWM three-frame	final	89.7	85.7	-4.0 [-7.3, -1.3]
Fast-LeWM	any	84.7	84.0	-0.7 [-1.7, +0.0]
Fast-LeWM	block	72.7	68.7	-4.0 [-6.3, -2.0]
Fast-LeWM	final	45.3	42.7	-2.7 [-4.7, -1.0]
AP	any	100.0	100.0	+0.0 [+0.0, +0.0]
AP	block	97.7	96.7	-1.0 [-2.3, +0.0]
AP	final	72.3	66.0	-6.3 [-9.3, -3.7]
AD-WM	any	86.2	84.3	-1.9 [-2.8, -1.1]
AD-WM	block	75.1	69.8	-5.3 [-7.1, -3.7]
AD-WM	final	52.6	47.8	-4.8 [-6.2, -3.4]

Table 26. Audit timeline for the Reacher correction. Times are UTC; hashes are the first 12 characters of each record's SHA–256. The launch authorization follows numerical gates, the sealed correction registration and verified local return.

Event	UTC	SHA–256
Earlier launch (later revoked)	2026-10-13 04:34:53.637935	`9163fb48cc4b`
Revocation; 33 records excluded	2026-10-13 05:51:03.922581	`0bc827769740`
Final numerical gates	2026-10-13 06:04:52.686207	`aebaa7ff7f6b`
Final correction registration	2026-10-13 06:06:33.085208	`5d82bd30efac`
Verified local return	2026-10-13 06:09:12.991794	`f046a7367651`
Formal launch authorization	2026-10-13 06:09:13.027389	`eb30a3c15e7a`
Corrected formal stage 1 started	2026-10-13 06:09:58.973668	`affcc5a3a0fa`
All corrected formal records complete	2026-10-13 07:27:50.541982	`f8198fe6659b`
Final return verified	2026-10-13 07:35:33.148483	`f918a16daba2`
Stop request accepted	2026-10-13 07:36:45.485577	`6f58cd00b6c2`

G Compute accounting

The formal runs used the recorded GPU-hour totals in the execution receipts. Compute accounting is reported for reproducibility and planning; it does not enter any success contrast. The analysis, figure, and table generators run on CPU from the machine-readable result files.

Table 27. Compute and reproducibility accounting. The correction is additional to the earlier allocation. GPU hours sum condition-process time; no model training was performed.

Item	Recorded value
Earlier formal evaluation	66.86 GPU h
Original held-out evaluation	8.36 GPU h
Earlier TECH/cache/reset work	2.31 GPU h (included in the 77.53 GPU h earlier total)
Corrected Reacher formal evaluation	2.529 GPU h
Original held-out and corrected-run hardware	Both 4 × RTX 4090; NVIDIA driver 595.84; CUDA 12.8
Experiment software	Python 3.12.3; stable-worldmodel commit `63988116`; full environment freeze in materials
Case bootstrap	5,000 replicates; frozen statistics function
This revision	CPU analysis of saved logs only; no new model or environment execution

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